



Rajiv Gandhi Proudhyogiki Vishwavidyalaya

Department of Information Technology

ACADEMIC PERFORMANCE REPORT

AI-Powered Comprehensive Analytics Report

PROGRAM

B.Tech.

SEMESTER

All Semesters

GENERATED

July 20, 2026 at 08:33 PM

JOB ID

JOB-RGPV_Result_0105IT2410_1-2

Generated by ResultAI — Automatic AI Report Generator

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Note: Page numbers are approximate and may vary based on data volume. This report contains AI-generated analytics and automatically computed insights.

EXECUTIVE SUMMARY

This report presents a comprehensive analysis of academic performance for 2 students enrolled in the **B.Tech.** program at **Rajiv Gandhi Proudhyogiki Vishwavidyalaya**. The analysis covers overall statistics, branch-wise comparisons, subject-level insights, grade distributions, risk assessments, and AI-generated recommendations for academic improvement.

KEY PERFORMANCE INDICATORS

2 Total Students	50.0% Pass Rate	50.0% Fail Rate	6.92 Avg SGPA
6.91 Avg CGPA	8.13 Highest SGPA	5.71 Lowest SGPA	6.92 Median SGPA

KEY FINDINGS

- **Moderate performance** — 50.0% pass rate indicates need for academic intervention programs.
- **Batch Topper:** AASTHA AGRAWAL (0105IT241001) achieved the highest SGPA of 8.13.
- The average SGPA of **6.92** is satisfactory but has scope for improvement.
- **Best performing department:** Information Technology.
- **Most challenging subject:** IT305- [T] recorded the highest failure rate.
- **1 student(s)** identified as academically at-risk and require targeted intervention.

PERFORMANCE SNAPSHOT

Metric	Value
Total Students Evaluated	2
Students Passed	1 (50.0%)
Students Failed	1 (50.0%)
Average SGPA	6.92
Median SGPA	6.92

Metric	Value
Standard Deviation	1.71
Average CGPA	6.91
Highest SGPA	8.13 (AASTHA AGRAWAL)
Lowest SGPA	5.71
Departments Analyzed	1
Subjects Analyzed	12

OVERALL STATISTICS

Complete statistical overview of 2 students across 1 department(s) and 12 subject(s).

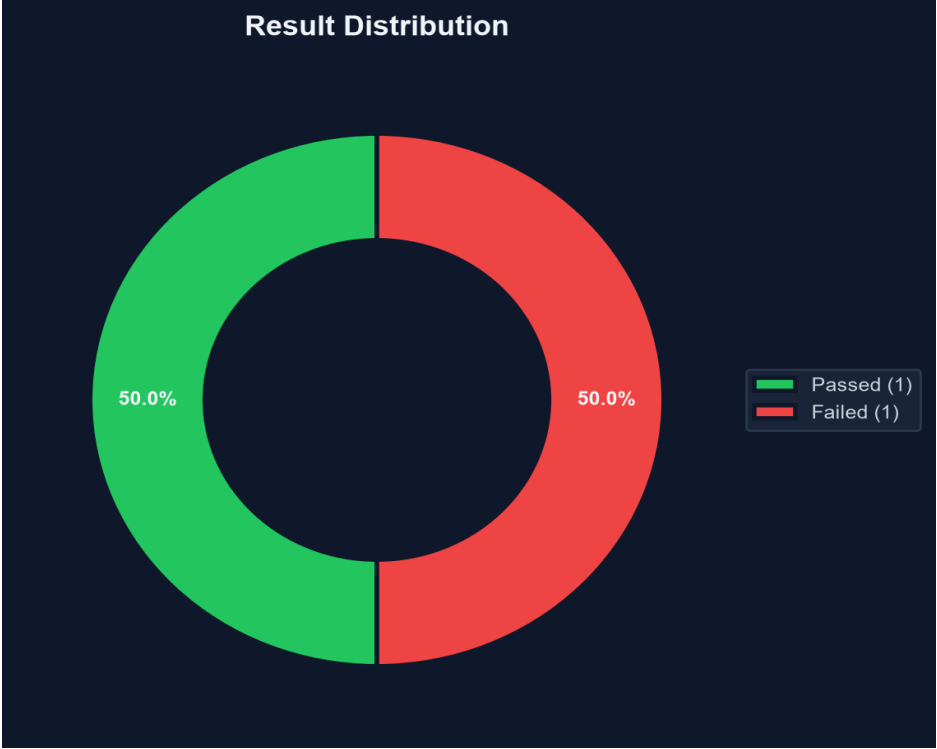
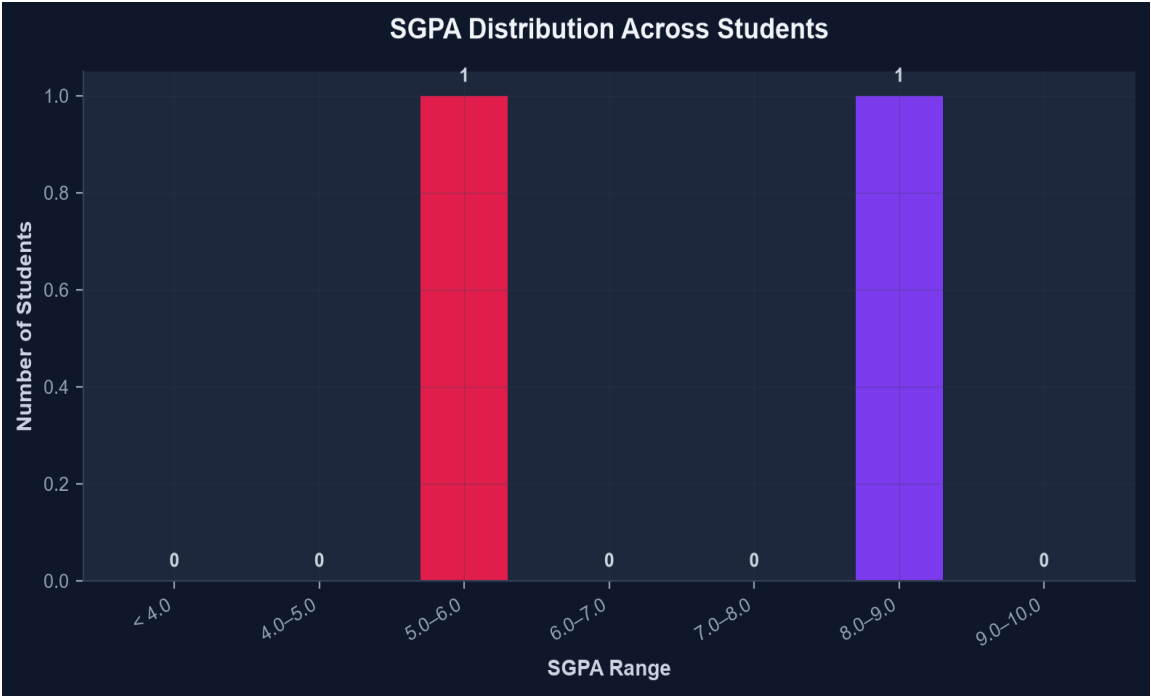
RESULT BREAKDOWN

Category	Count	Percentage
Total Students	2	100%
Passed	1	50.0%
Failed	1	50.0%
Unknown/Pending	0	0.0%

SGPA STATISTICS

Statistic	SGPA	CGPA
Highest	8.13	8.18
Lowest	5.71	5.64
Average (Mean)	6.92	6.91
Median	6.92	6.91
Std. Deviation	1.71	—

SGPA DISTRIBUTION



BRANCH-WISE ANALYSIS

Comparative performance analysis across 1 academic department(s). This section evaluates each branch based on average SGPA, pass rate, failure rate, and backlog distribution.

DEPARTMENT COMPARISON TABLE

Branch	Students	Pass %	Fail %	Avg SGPA	Avg CGPA	Backlogs	Topper
IT	2	50.0%	50.0%	6.92	6.91	1	AASTHA AGRAWAL

SEMESTER ANALYSIS

Performance analysis for **Semester Current** of the **B.Tech.** program. This section provides a snapshot of the semester's academic outcomes.

SEMESTER PERFORMANCE OVERVIEW

Metric	Value
Semester	Semester Current
Program	B.Tech.
Total Students	2
Pass Count	1
Fail Count	1
Pass Percentage	50.0%
Average SGPA	6.92
Average CGPA	6.91
Highest SGPA	8.13
Lowest SGPA	5.71
Number of Subjects	12
Departments	1

DEPARTMENT-WISE SEMESTER RESULTS

Department	Students	Pass %	Avg SGPA	Topper
Information Technology	2	50.0%	6.92	AASTHA AGRAWAL

Note: Multi-semester trend analysis requires comparison data from previous semesters. Use the --compare flag with a previous semester's Excel file to enable trend charts.

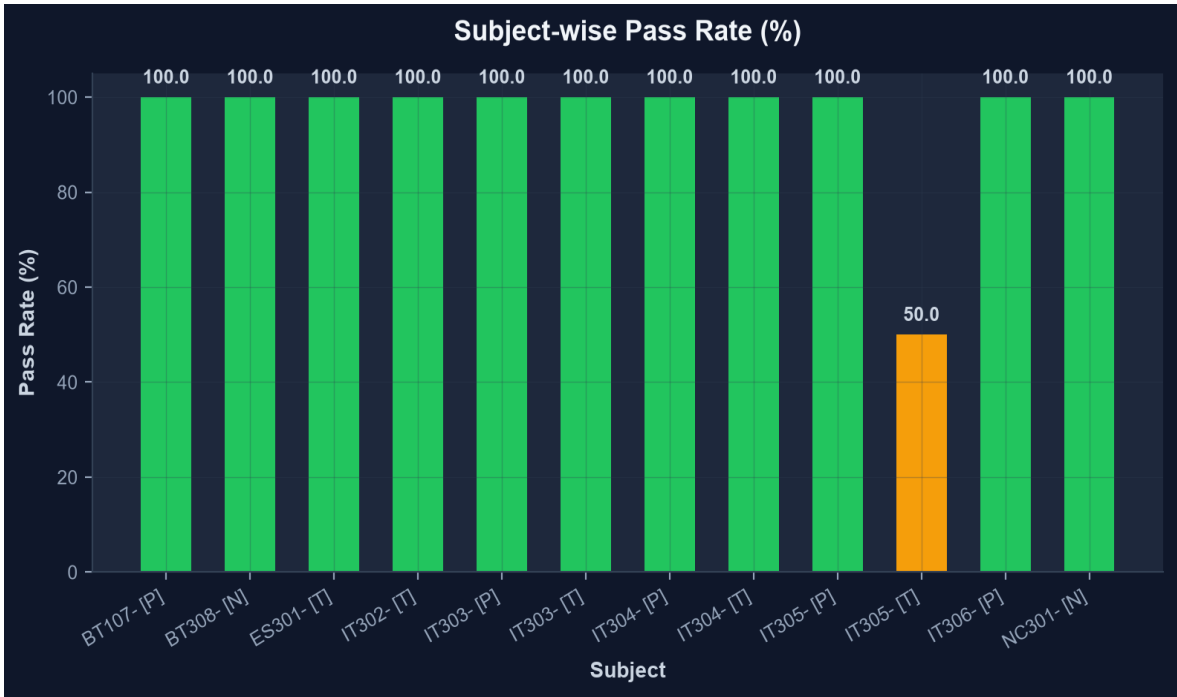
SUBJECT-WISE ANALYSIS

Detailed analysis of **12** subject(s) evaluated in this semester. Each subject is assessed for pass rate, failure rate, average grade, difficulty index, and grade distribution.

SUBJECT PERFORMANCE SUMMARY

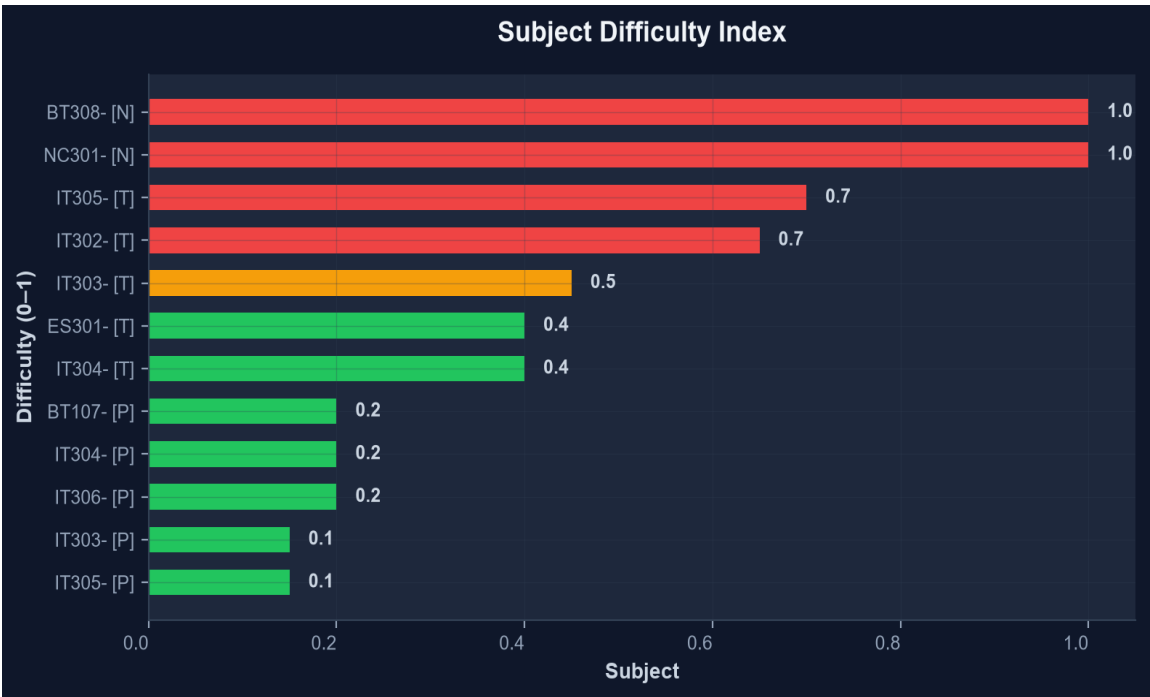
Subject	Students	Pass %	Fail %	Avg GP	Difficulty	Highest	Lowest
IT305- [T]	2	50.0%	50.0%	3.0	0.7	B	F
BT107- [P]	2	100.0%	0.0%	8.0	0.2	A	A
BT308- [N]	2	100.0%	0.0%	0.0	1.0	—	—
ES301- [T]	2	100.0%	0.0%	6.0	0.4	B+	C
IT302- [T]	2	100.0%	0.0%	3.5	0.65	B+	B+
IT303- [P]	2	100.0%	0.0%	8.5	0.15	A+	A
IT303- [T]	2	100.0%	0.0%	5.5	0.45	B	C
IT304- [P]	2	100.0%	0.0%	8.0	0.2	A	A
IT304- [T]	2	100.0%	0.0%	6.0	0.4	B+	C
IT305- [P]	2	100.0%	0.0%	8.5	0.15	A+	A
IT306- [P]	2	100.0%	0.0%	8.0	0.2	A	A
NC301- [N]	2	100.0%	0.0%	0.0	1.0	—	—

SUBJECT-WISE PASS RATE



SUBJECT DIFFICULTY INDEX

The Difficulty Index ranges from 0.0 (easiest) to 1.0 (hardest), calculated based on average grade points. Higher values indicate more challenging subjects.



DETAILED SUBJECT BREAKDOWN

- **BT107- [P]:** 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.0, Failed: 0 — Grades: [A: 2]
- **BT308- [N]:** 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 0.0, Failed: 0 — Grades: [NA: 2]
- **ES301- [T]:** 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 6.0, Failed: 0 — Grades: [B+: 1, C: 1]
- **IT302- [T]:** 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 3.5, Failed: 0 — Grades: [B+: 1, C+: 1]
- **IT303- [P]:** 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.5, Failed: 0 — Grades: [A: 1, A+: 1]
- **IT303- [T]:** 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 5.5, Failed: 0 — Grades: [B: 1, C: 1]
- **IT304- [P]:** 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.0, Failed: 0 — Grades: [A: 2]
- **IT304- [T]:** 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 6.0, Failed: 0 — Grades: [B+: 1, C: 1]
- **IT305- [P]:** 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.5, Failed: 0 — Grades: [A: 1, A+: 1]

- **IT305- [T]**: 2 students, Pass Rate: 50.0%, Fail Rate: 50.0%, Avg Grade Point: 3.0, Failed: 1 — Grades: [B: 1, F: 1]
- **IT306- [P]**: 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.0, Failed: 0 — Grades: [A: 2]
- **NC301- [N]**: 2 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 0.0, Failed: 0 — Grades: [NA: 2]

Most Failed Subject: IT305- [T]

Easiest Subject: IT303- [P] (Difficulty Index: 0.15)

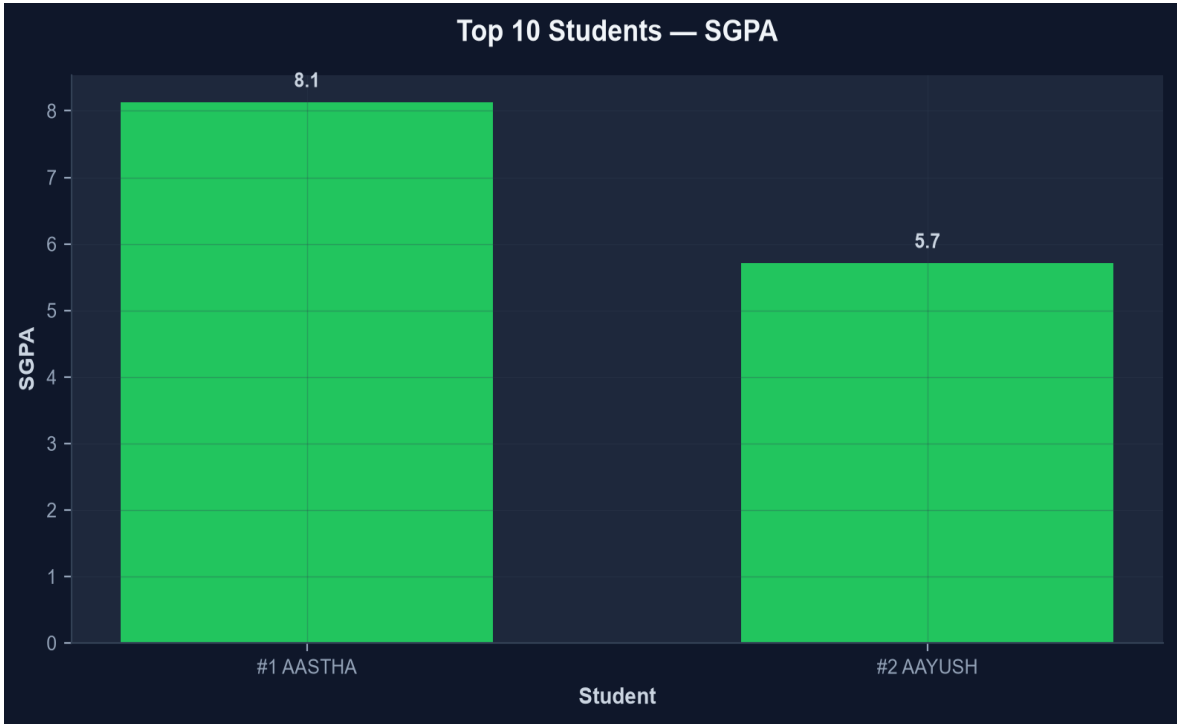
TOPPER ANALYSIS

Academic rank list of top-performing students. Rankings are based on Semester GPA (SGPA), with CGPA as a tiebreaker.

TOP 10 STUDENTS

Rank	Enrollment	Name	Branch	SGPA	CGPA	Result
1st	0105IT241001	AASTHA AGRAWAL	IT	8.13	8.18	PASS
2nd	0105IT241002	AAYUSH MEHRA	IT	5.71	5.64	FAIL

TOP STUDENTS — SGPA VISUALIZATION



STUDENT PERFORMANCE DETAILS

Individual performance cards for 2 student(s). Each card shows subject grades, SGPA, CGPA, result status, and backlog count.

#1 AASTHA AGRAWAL

0105IT241001

Branch: Information Technology | Program: B.Tech. | Semester: N/A

SUBJECT CODE	GRADE
ES301- [T]	B+
IT302- [T]	B+
IT303- [T]	B
IT304- [T]	B+
IT305- [T]	B
BT107- [P]	A
IT303- [P]	A+
IT304- [P]	A
IT305- [P]	A+
IT306- [P]	A
BT308- [N]	NA
NC301- [N]	NA

SGPA

8.13

CGPA

8.18

RESULT

PASS

BACKLOGS

0

#2 AAYUSH MEHRA

0105IT241002

Branch: Information Technology | Program: B.Tech. | Semester: N/A

SUBJECT CODE	GRADE
ES301- [T]	C
IT302- [T]	C+
IT303- [T]	C
IT304- [T]	C
IT305- [T]	F
BT107- [P]	A
IT303- [P]	A
IT304- [P]	A
IT305- [P]	A
IT306- [P]	A
BT308- [N]	NA
NC301- [N]	NA

SGPA

5.71

CGPA

5.64

RESULT

FAIL

BACKLOGS

1



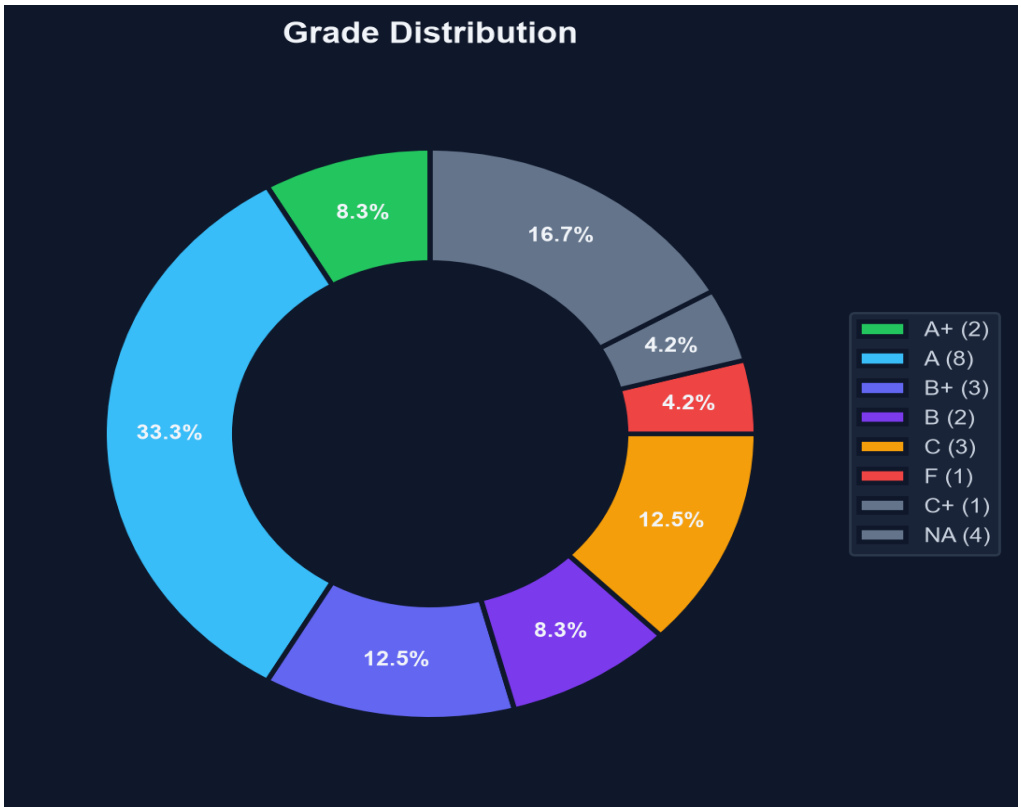
GRADE DISTRIBUTION

Distribution analysis of 24 total grades awarded across all subjects and students.

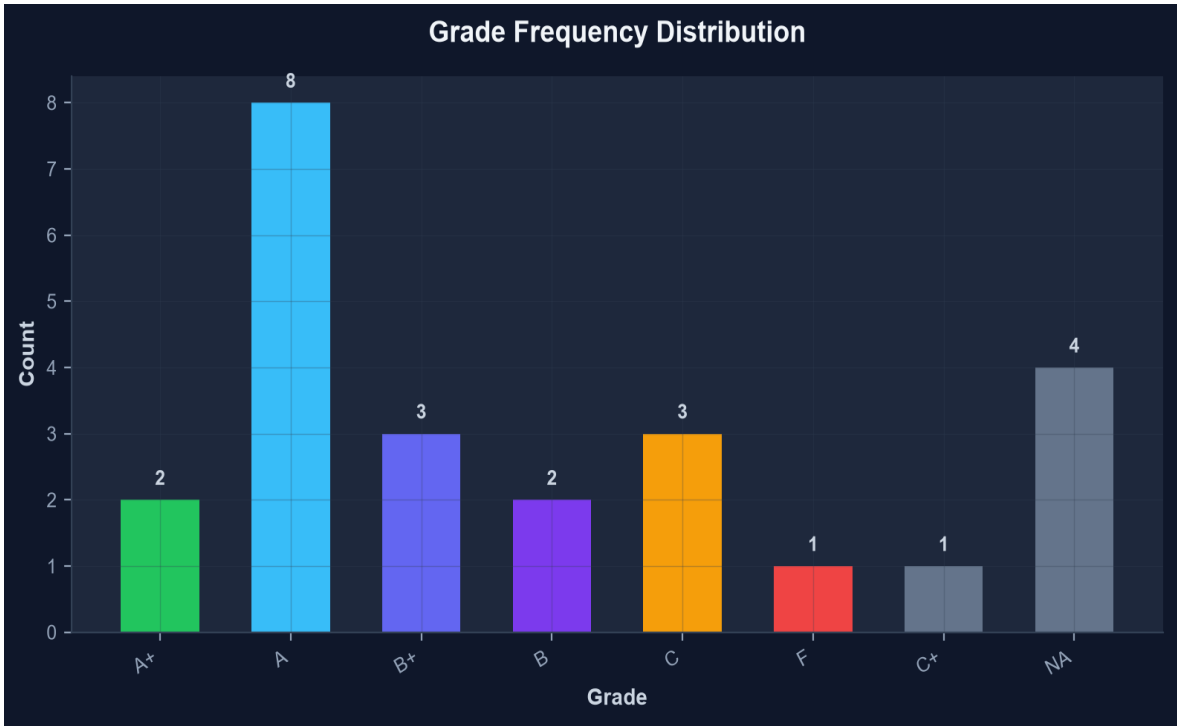
GRADE FREQUENCY TABLE

Grade	Count	Percentage	Performance Level
A+	2	8.3%	Excellent
A	8	33.3%	Very Good
B+	3	12.5%	Good
B	2	8.3%	Above Average
C	3	12.5%	Average
F	1	4.2%	Fail
C+	1	4.2%	C+
NA	4	16.7%	NA

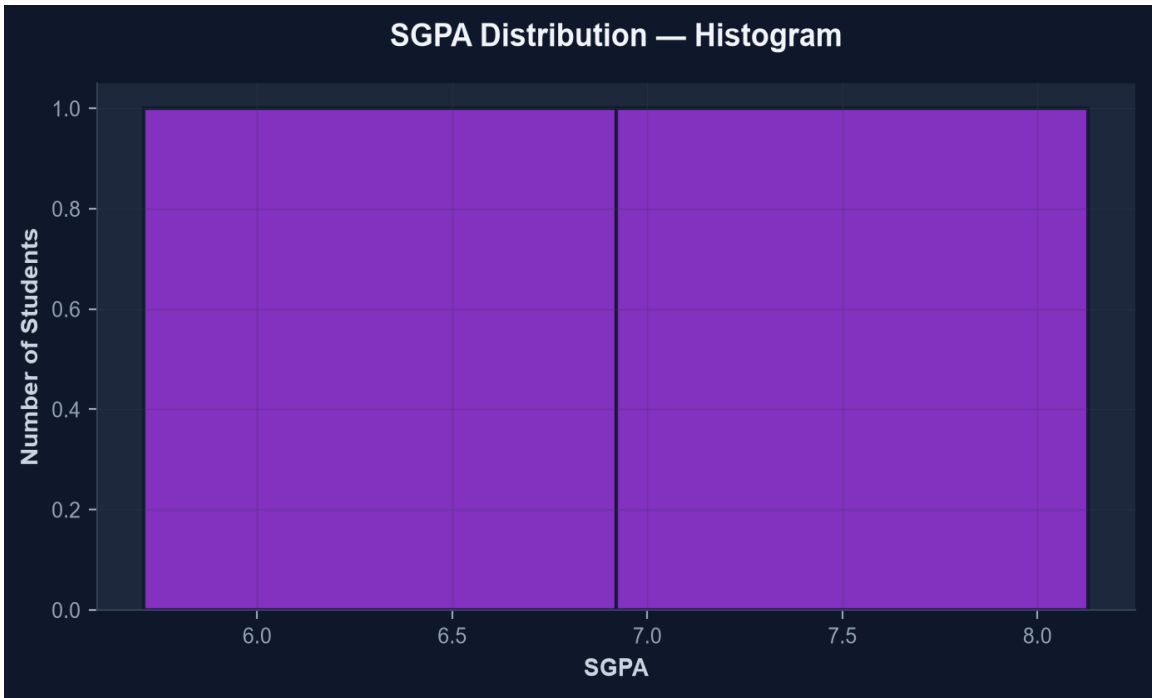
GRADE DISTRIBUTION — DONUT CHART



GRADE DISTRIBUTION — BAR CHART



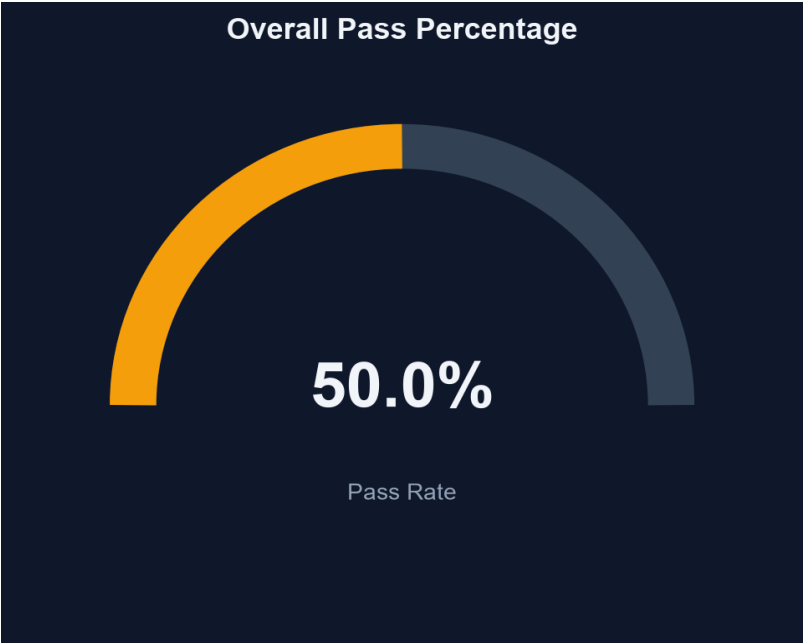
SGPA FREQUENCY HISTOGRAM



PASS PERCENTAGE ANALYSIS

Overall pass rate analysis showing the proportion of students who cleared all subjects versus those with one or more failures.

OVERALL PASS RATE GAUGE



DEPARTMENT-WISE PASS RATE

Department	Total	Passed	Failed	Pass %
Information Technology	2	1	1	50.0%



BACKLOG ANALYSIS

Analysis of 1 total backlog instances across 1 student(s). Backlogs include F, Fail, AB, Absent, and RL grades.

DEPARTMENT-WISE BACKLOG DISTRIBUTION

Department	Total Backlogs
Information Technology	1



SUBJECT-WISE BACKLOG DISTRIBUTION

Subject Code	Failure Count
IT305	1



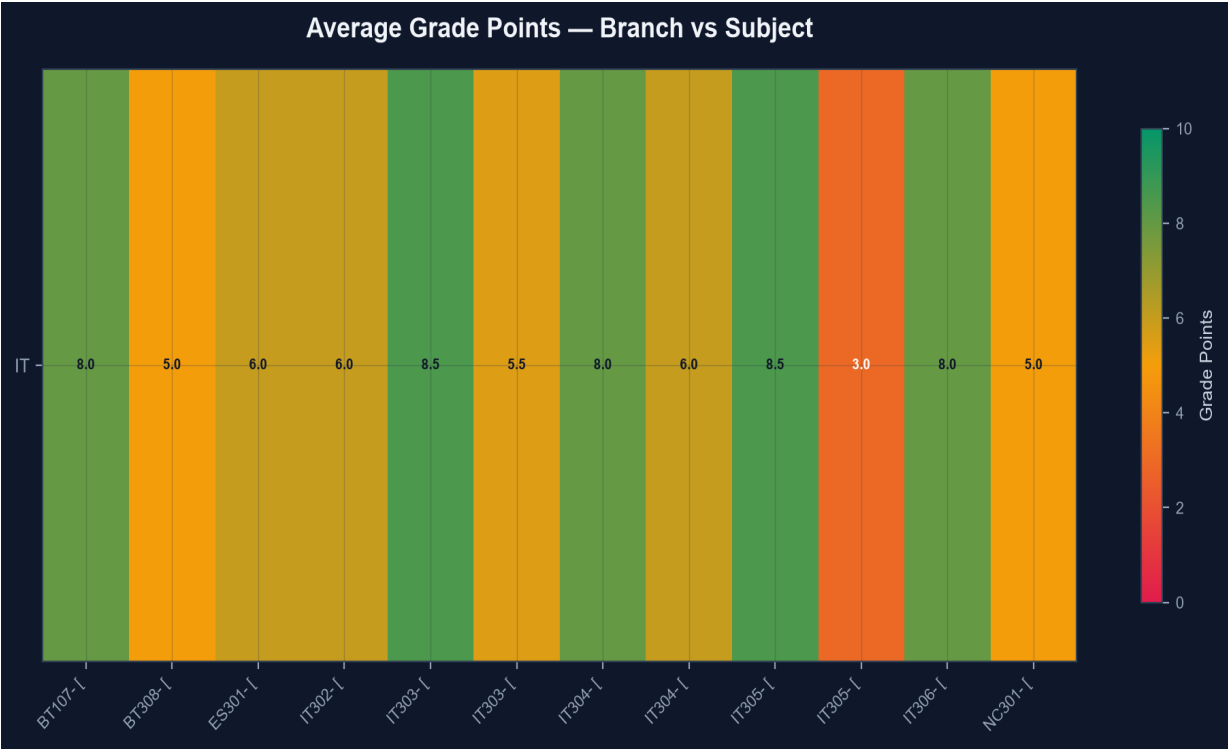
STUDENTS WITH BACKLOGS

Enrollment	Name	Branch	SGPA	Backlogs	Failed Subjects
0105IT241002	AAYUSH MEHRA	IT	5.71	1	IT305

PERFORMANCE HEATMAPS

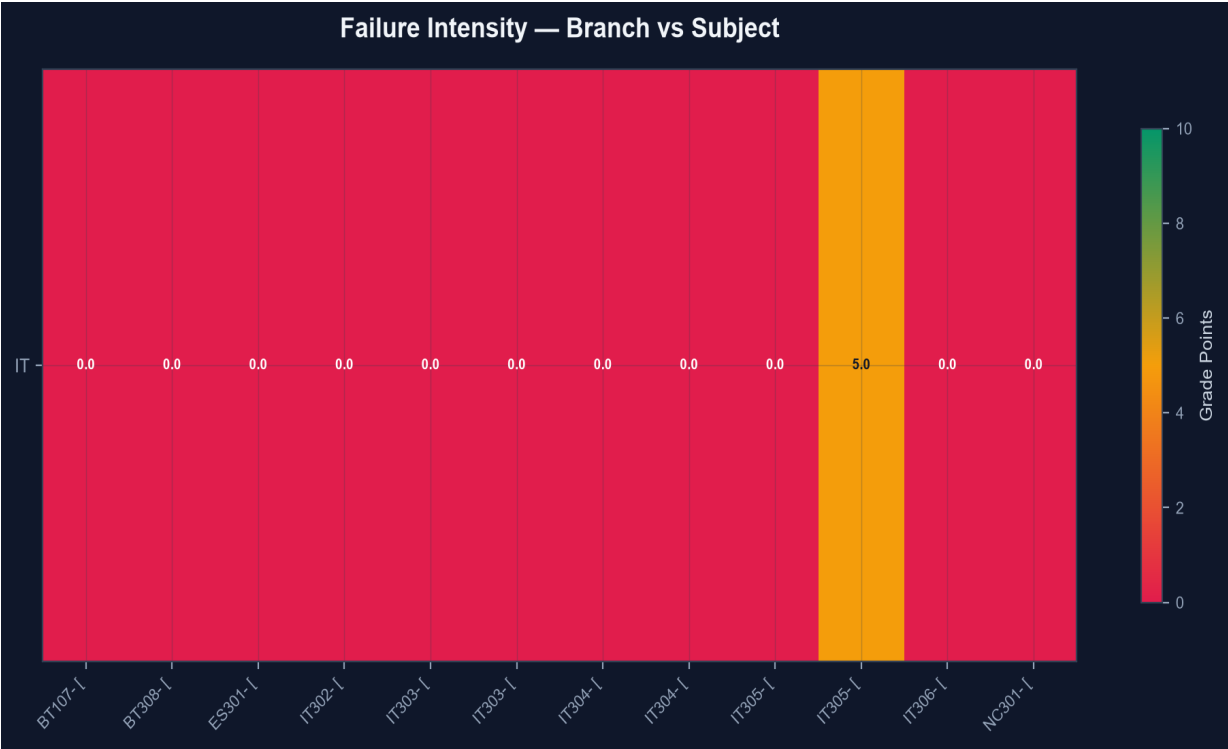
Visual heatmaps showing performance intensity across branches and subjects. Warmer colors (red/orange) indicate weaker performance; cooler colors (green) indicate stronger results. Grade points range from 0 (Fail) to 10 (Outstanding).

BRANCH × SUBJECT PERFORMANCE HEATMAP



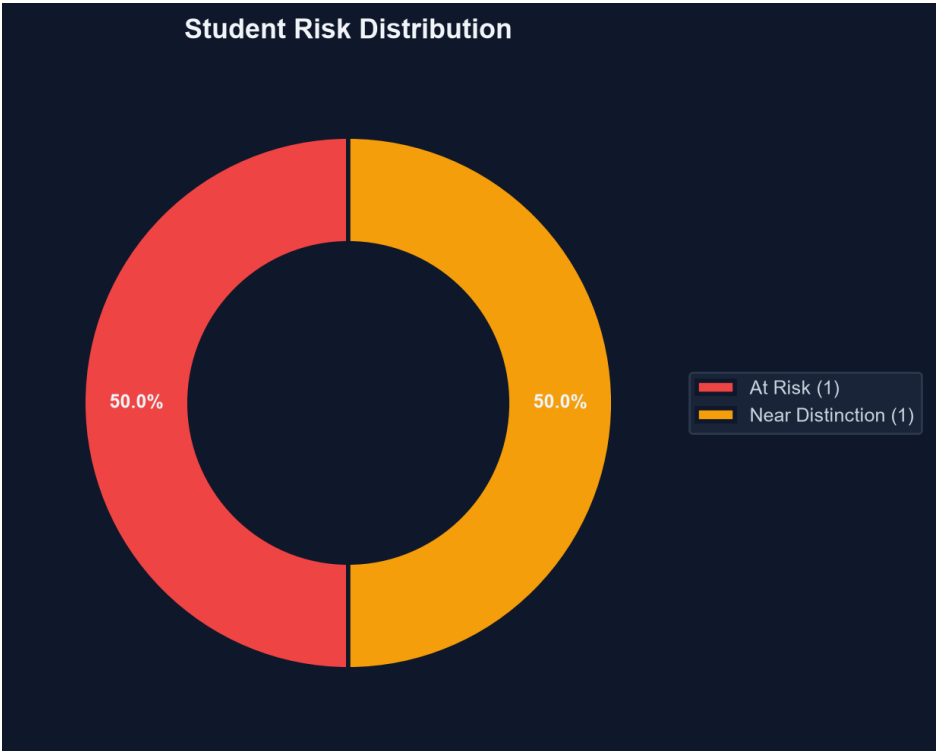
FAILURE INTENSITY HEATMAP

Higher values (red) indicate higher failure rates. Scale: 0 = No Failures, 10 = 100% Failure.



RISK ANALYSIS

Academic risk stratification identifying students who need attention. **1** at-risk, **1** near-distinction, **0** with multiple backlogs.



STUDENTS AT RISK (SGPA < 5.0 or FAIL status)

Enrollment	Name	Branch	SGPA	CGPA	Backlogs
0105IT241002	AAYUSH MEHRA	Information Tec	5.71	5.64	1

STUDENTS NEAR DISTINCTION (SGPA 7.5–8.5)

Enrollment	Name	Branch	SGPA	CGPA
0105IT241001	AASTHA AGRAWAL	Information Tec	8.13	8.18

IMPROVEMENT ANALYSIS

Improvement analysis requires comparison data from a previous semester or job. To enable this section, generate the report with the **--compare** flag pointing to a previous result Excel file.

Example: `python -m report_generator.generate_report --input current.xlsx --compare previous.xlsx`

COMPARISON ANALYSIS

Comparison analysis is available when a previous dataset is provided for reference. Use the **--compare** flag to specify a previous result Excel file.

Example: `python -m report_generator.generate_report --input current.xlsx --compare previous.xlsx`

AI-GENERATED INSIGHTS

The AI analytics engine has generated **34** observations based on comprehensive statistical analysis of the academic data. These insights are automatically derived from patterns, distributions, and trends.

- 1

A total of 2 students were analyzed in this academic evaluation.
- 2

Overall pass percentage stands at 50.0%, with 1 students clearing all subjects.
- 3

Failure rate is recorded at 50.0%, affecting 1 students.
- 4

Moderate pass rate indicates significant academic challenges that need attention.
- 5

Average SGPA across all students is 6.92 out of 10.0.
- 6

Median SGPA is 6.92, indicating the central tendency of student performance.
- 7

Highest SGPA achieved is 8.13 by AASTHA AGRAWAL (0105IT241001).
- 8

Lowest SGPA recorded is 5.71.
- 9

Standard deviation of SGPA is 1.71, measuring the spread of academic performance.
- 10

Average Cumulative GPA (CGPA) is 6.91.
- 11

Information Technology is the best-performing department based on average SGPA.
- 12

Information Technology is the weakest-performing department and may need academic support.
- 13

Information Technology department: 2 students, 50.0% pass rate, average SGPA 6.92.
- 14

Information Technology recorded 1 total subject backlogs.
- 15

Most failed subject: IT305- [T] — requires focused remedial action.
- 16

Subject BT308- [N] has a high difficulty index of 1.0 (scale 0-1).
- 17

Subject IT302- [T] has a high difficulty index of 0.65 (scale 0-1).
- 18

Subject IT305- [T] has a high failure rate of 50.0% — classified as a difficult subject.

-
- 19 Subject IT305- [T] has a high difficulty index of 0.7 (scale 0-1).
-
- 20 Subject NC301- [N] has a high difficulty index of 1.0 (scale 0-1).
-

-
- 21 1 student(s) identified as academically at-risk (SGPA < 5.0 or FAIL status).
-
- 22 1 student(s) are near distinction level (SGPA 7.5–8.5) — targeted support could push them higher.
-
- 23 'A+' grade was the achieved 2 times (8.3% of all grades).
-
- 24 1 'F' (fail) grades recorded (4.2% of all grades), each representing a subject backlog.
-
- 25 1 students (50.0%) scored SGPA above 8.0, demonstrating strong academic excellence.
-
- 26 Batch topper is AASTHA AGRAWAL (0105IT241001) with SGPA 8.13.
-
- 27 This report covers a cohort of 2 students from the AASTHA batch.
-
- 28 Out of 2 students evaluated, 1 successfully cleared all subjects.
-
- 29 Academic performance data has been analyzed across multiple dimensions including branch, subject, and individual student levels.
-
- 30 Statistical measures including mean, median, and standard deviation have been computed for comprehensive analysis.
-
- 31 Grade distribution analysis reveals the spread of academic achievement across all evaluated subjects.
-
- 32 Backlog analysis identifies subjects and departments requiring targeted academic interventions.
-
- 33 Risk stratification categorizes students based on their likelihood of academic difficulty.
-
- 34 This AI-generated report provides data-driven insights for informed academic decision-making.
-

AI RECOMMENDATIONS

The AI engine has generated **19** actionable recommendations based on the analysis findings. Recommendations are prioritized as **HIGH**, **MEDIUM**, and **LOW** priority.

Priority Distribution: **3 HIGH** | **8 MEDIUM** | **8 LOW**

HIGH	R1	Urgent: 50.0% failure rate detected. Conduct an academic review board meeting to identify root causes and implement immediate corrective measures.
HIGH	R2	Conduct remedial classes and extra tutorials for IT305- [T] — failure rate of 50.0% indicates significant difficulty.
HIGH	R3	Identify and assign academic mentors to 1 at-risk student(s) immediately.
MEDIUM	R4	Review curriculum difficulty for BT308- [N] (difficulty index: 1.0). Consider adjusting teaching methodology or evaluation pattern.
MEDIUM	R5	Review curriculum difficulty for NC301- [N] (difficulty index: 1.0). Consider adjusting teaching methodology or evaluation pattern.
MEDIUM	R6	Schedule one-on-one counseling sessions with all at-risk students to understand personal/academic challenges.
MEDIUM	R7	Provide targeted coaching to 1 near-distinction student(s) — small improvement could push them to distinction level.
MEDIUM	R8	Increase practical and hands-on laboratory sessions to improve conceptual understanding.
MEDIUM	R9	Implement continuous internal assessment to provide early warning signals for academic difficulties.
MEDIUM	R10	Organize industry expert guest lectures to improve student engagement and practical understanding.
MEDIUM	R11	Implement a digital learning management system for supplementary resources and self-paced revision.

LOW

R12 Recognize and reward top-performing students with merit certificates and academic incentives.

LOW

R13 Encourage toppers to participate in peer tutoring programs to uplift overall batch performance.

LOW

R14 Conduct student feedback surveys on teaching quality and curriculum relevance.

LOW

R15 Establish study groups and collaborative learning environments for improved peer learning.

- LOW

R16

Review and update course materials to align with current industry standards and practices.
- LOW

R17

Organize academic orientation sessions at the start of each semester to set performance expectations.
- LOW

R18

Publish this analytics report to department heads and academic coordinators for data-driven planning.
- LOW

R19

Schedule regular parent-teacher meetings for students with declining performance trends.